Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/07/2025
PI Name:	Macaulay, Timothy Ph.D.		
Project Title:	Countermeasures for Mitigation of Sensorimotor Decrements Following Head-Down Bed Rest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3711	**Congressional District:**	36
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
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Task Description:	Note: Research aims merged with Grant #80NSSC21K0369 (PI: Rutkove). Astronauts experience post-landing disturbances in postural and locomotor control due to sensorimotor adaptations during spaceflight. During exploration missions, these alterations could delay operations, impact the completion of mission milestones, and increase crew injury risk due to falls. Although exercise is effective for mitigating cardiovascular and muscular deconditioning, new approaches are needed to further preserve sensorimotor function, especially during and after exploration missions with less comprehensive exercise capabilities than the current suite on the International Space Station (ISS). We have identified proprioceptive training and electrical muscle stimulation (EMS) as promising in-flight countermeasures. Since prolonged head down bed rest (HDBR) is a spaceflight analog for body unloading and causes postural and locomotor control decrements that parallel those observed after spaceflight, it can be used to accelerate the development of these countermeasures. This study will determine the effects of proprioceptive training and EMS on functional task performance and sensorimotor function following 60 days of 6° HDBR. We will compare four groups of subjects – an EMS arm, a proprioceptive training arm, an exercise plus proprioceptive training arm, and a control arm. The EMS countermeasure will include daily bilateral stimulation of lower extremity muscles (30 minutes per session). Proprioceptive training will be performed three days per week (30 minutes per session) consisting of body-loaded postural tasks in the horizontal position on an air bearing sled. This training targets the activation and control of small postural muscles to keep the proprioceptive and tactile systems conditioned and adaptable to upright balance challenges. Horizontal exercise training will mimic current protocols used on the ISS, but treadmill aerobic exercise will be replaced with additional cycling aerobic exercise. Primary outcome measures will include functional tests that are representative of high priority exploration mission tasks and require high demand for dynamic control of postural stability. Additional measures will be used to identify the key physiological factors contributing to countermeasure benefits, including assessments of sensorimotor function, somatosensory feedback, musculoskeletal function, muscle structure using MRI, ultrasound-guided muscle thickness and echointensity, electrical impedance myography, and DXA bone scans. Additionally, serological measurements, neuromuscular biomarker and circulating miRNAs, will be investigated for their potency to be used as biomarkers of muscle degeneration and indicators of treatment response. Associations between these changes in key physiological factors and changes in functional task performance will be explored. Additional measures have been added through a study augmentation to collect and analyze outcome measures needed to assess Spaceflight Associated Neuro-ocular Syndrome (SANS) and internal jugular vein (IJV) flow stasis and thrombus formation. The Cardiovascular and Vision Laboratory (CVL) first identified the presence of a thrombus in the IJV of a crewmember flying a long-duration mission to the ISS. Concurrently, the CVL team developed and implemented a scoring system to assess venous flow patterns in the IJV, documenting flow stasis in the left IJV in some, but not all, crewmembers; flow stasis represents a risk factor for thrombus development. Therefore, there would be value in determining if a spaceflight analog can be used to (1) identify if flow stasis develops that would pose a risk for thrombus formation, (2) determine whether IJV blood flow hemodynamics are related to the development of SANS outcomes, and (3) further characterize SANS findings including brain structural changes and assessment of retinal vascular leakage. The strict head-down tilt analog environment is a spaceflight analog that induces SANS in a subset of participants and facilitates the development of protocols and approaches to investigate SANS and associated physiological changes. The CVL assessed retinal vascular leakage in a previous bed rest campaign (SANS Countermeasure Bed Rest) using intravenous fluorescein angiography (IVFA) to determine whether optic disc edema in SANS results in leaky retinal vasculature. Preliminary data revealed leakage ~10 minutes after dye injection in those presenting with optic disc edema, but not in those who did not develop edema. These preliminary findings warrant further investigation into the timing of vascular leakage during IVFA imaging, and the potential consequence of “late leakage”.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Exploration missions must be accomplished under new constraints and fewer exercise hardware capabilities. New countermeasures are needed for these missions to replace cumulative sensorimotor stimuli and must be developed in collaboration with other NASA stakeholders. We have designed a countermeasure system to both train and assess proprioceptive function in-flight and are working the Astronaut Strength, Conditioning, and Rehabilitation (ASCR) group and members of the Human Physiology, Performance, Protection, & Operations (H-3PO) Laboratory to define the protocol. The project must be conducted in a spaceflight analog due to testing limitations on the International Space Station (ISS) and the need to determine efficacy in a controlled environment. Therefore, this research effort will require leveraging inhouse expertise including experience with head-down bed rest (HDBR) models to evaluate countermeasures. Outcomes will include functional tests previously established by Johnson Space Center (JSC) laboratories that represent critical mission tasks and have demonstrated performance decrements after both spaceflight and HDBR. It is also essential to track performance and understand the underlying factors that contribute to efficacy in order to translate the countermeasure to spaceflight operations and predict outcomes during exploration missions. The investigators of this research proposal have extensive experience conducting HDBR studies and a strong working relationship with other NASA stakeholders. Proprioceptive training is a novel approach, but a team of sensorimotor and rehabilitation experts to collaboratively define and test the countermeasure increases the likelihood of study success.

Research Impact/Earth Benefits:	The deliverable from this project will be proof-of-concept countermeasure designs for maintaining functional task performance with full assessment of efficacy in a spaceflight analog. If the countermeasures are effective, they will be translated for evaluation with the suite of operationally implemented in-flight ISS and exploration countermeasures. This will help mitigate the risk of altered sensorimotor function impacting critical mission tasks. The inclusion of various key physiological measures will also lead to a greater fundamental understanding of how proprioceptive, tactile, and muscular functions contribute to upright functional mobility. These findings may translate to ground-based interventions for various clinical populations that require balance training and fall risk mitigation. The added SANS measures will help further characterize the frequency and magnitude of vascular leakage associated with optic disc edema, venous blood flow and anatomical variability, and the brain structural changes.
Task Progress:	Campaign 1 was completed in December 2024. One participant was dropped from the study early in head-down tilt (approximately HDT2) due to incidental findings in their brain MRI. This finding was unfortunately too late for enrolling one of the backup participants. Therefore, 11 participants completed Campaign 1. The team decided to move one Control subject to the Exercise plus Proprioception Training (EPT) group so that all countermeasure groups had 3 participants, and the Control group had 2 participants. During HDT, one participant in the EPT group missed a few exercise countermeasure sessions due to motion sickness-like symptoms. However, the symptoms subsided with mild treatments, and the subject completed all remaining countermeasure and data collection sessions. All post-HDBR (R+) testing was completed nominally despite some minor foot sole discomfort being reported by one subject. Data from functional performance tests and proprioception function tests were processed for preliminary presentations to program management (e.g., NASA Human Research Program/HRP and Mars Campaign Office). Campaign 2 began in April 2025. Prior to the start, additional questionnaires were created to better characterize some of the findings in Campaign 1. These were added to the NASA Institutional Review Board (IRB), as well as the updates made to the integrated requirements document, data collection overview, and daily test schedule. In addition, maintenance functions were performed, such as shipping spare parts and (re)training German Aerospace Center (DLR) operators. The DLR team successfully completed Campaign 2 with 12 HDBR participants. One subject reported pain during active dorsiflexion movements during proprioception testing on HDT59. Therefore, single-leg (right) proprioception assessments were skipped for that subject. However, the subject and DLR team communicated that the injury was not due to the device itself, but rather due to deconditioning, which is a known risk of long-duration HDBR. This was reported to the NASA IRB. The subject was able to successfully complete all post-HDBR (R+) testing. One other subject was released early (R+6) at their request due to personal reasons (unrelated to the study). Sufficient notice was given to adjust the testing schedule and move several data collections up to R+5 and R+6, thus minimizing science impacts. The subject was provided with their medical information and an explanation of the risks of leaving the study prior to full rehabilitation. Campaign 2 data processing is ongoing. The Cardiovascular and Vision Laboratory (CVL) received all anticipated datasets collected by DLR, including MRI (BDC-3 and HDT60), optical coherence tomography (OCT) (BDC-3, HDT28, R+0), intravenous fluoresceine angiography (IVFA) (R+3), and ultrasound (USND) cross sectional area and flow scores (BDC-9, HDT2, HDT15, HDT30, HDT45, HDT60, R+0). Data that will be analyzed by external collaborators in Fiscal Year 2026 has been sent to the respective teams (IVFA by Dr. Alex Huang and MRI by Dr. Larry Kramer). Data analyzed by the CVL (OCT and USND) have been completed for Campaign 1 and are ongoing for Campaign 2. Posturography data from previous HDBR studies were analyzed as part of the HRP Investigators' Workshop (IWS) Student Augmentation Award. Previously, we reported novel analyses on eight different outcome measures from 47 long-duration (≥60 days) HDBR subjects. Receiver Operating Characteristics (ROC) curves were generated for each outcome measure. Most outcome measures performed similarly in terms of area under the curve (AUC), with Minimum Time to Contact (mTTC) being the best measure. Therefore, this mTTC measure will be used for analyses of the Posturography data generated by the current HDBR study. In a separate analysis, we sought to determine how Posturography outcome measures are impacted by an accurate anthropometry-based center of mass (COM) estimate compared to the assumed 55% height that is typically used as the COM in the literature. Data from 30 subjects (15 women, 15 men) were used. Body scan data were collected using a Vitus 3D Body Scanner and were reposed to simulate arms across the chest. Sex and height were both significant predictors of COM %height, while weight was not. Men had significantly higher COM %height (58.1%) than women (56.8%). Posturography scores changed by less than 1% for 27 of the subjects when using the subjects' accurate COM %height rather than the 55% estimate. In addition, there was no clear reason/trend for the slightly larger changes in the remaining 3 subjects. Therefore, we determined that the impacts of using the subjects' accurate COM %height were significant, but not clinically meaningful.
Bibliography Type:	Description: (Last Updated: 12/23/2025)
Abstracts for Journals and Proceedings	Gustafson HE, Moudy SC, Macaulay TR, Wood SJ. "Balance performance after head-down bed rest: key outcome measures and physical characteristics." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Presentation. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Macaulay TR, Rutkove SB, Bent LR, Macias BR, Valero-Cuevas FJ, Oddsson L, Lawrence EL, Strock N, Youg M, Bloomberg JJ, Mortreux M, Bouxsein ML, Beheshti A, Abitante T, Newman D. Dunlap K, Sawyer A, Huang AS, Kramer L, Laurie SS, Wood SJ, Peters BT. "Countermeasures for mitigation of sensorimotor decrement following head-down tilt bed rest." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Presentation. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025